

Figure 5: Top original images, Bottom: reconstructions from a 2 stage VQ-VAE, with 3 latents to model the whole image (27 bits), and as such the model cannot reconstruct the images perfectly. The reconstructions are generated by sampled from the second PixelCNN prior in the 21x21 latent domain of first VQ-VAE, and then decoded with standard VQ-VAE decoder to 84x84. A lot of the original scene, including textures, room layout and nearby walls remain, but the model does not try to store the pixel values themselves, which means the textures are generated procedurally by the PixelCNN.

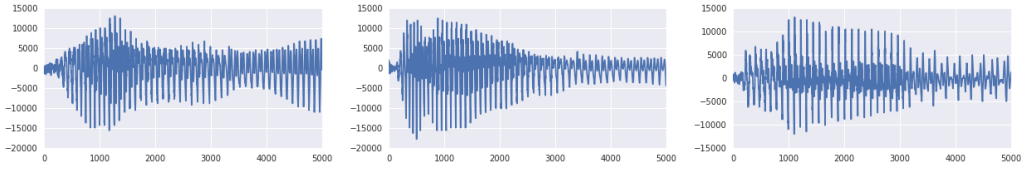


Figure 6: Left: original waveform, middle: reconstructed with same speaker-id, right: reconstructed with different speaker-id. The contents of the three waveforms are the same.

4.3 Audio

In this set of experiments we evaluate the behaviour of discrete latent variables on models of raw audio. In all our audio experiments, we train a VQ-VAE that has a dilated convolutional architecture similar to WaveNet decoder. All samples for this section can be played from the following url: <https://avdnoord.github.io/homepage/vqvae/>.

We first consider the VCTK dataset, which has speech recordings of 109 different speakers [41]. We train a VQ-VAE where the encoder has 6 strided convolutions with stride 2 and window-size 4. This yields a latent space 64x smaller than the original waveform. The latents consist of one feature map and the discrete space is 512-dimensional. The decoder is conditioned on both the latents and a one-hot embedding for the speaker.

First, we ran an experiment to show that VQ-VAE can extract a latent space that only conserves long-term relevant information. After training the model, given an audio example, we can encode it to the discrete latent representation, and reconstruct by sampling from the decoder. Because the dimensionality of the discrete representation is 64 times smaller, the original sample cannot be perfectly reconstructed sample by sample. As it can be heard from the provided samples, and as shown in Figure 7, the reconstruction has the same content (same text contents), but the waveform is quite different and prosody in the voice is altered. This means that the VQ-VAE has, without any form of linguistic supervision, learned a high-level abstract space that is invariant to low-level features and only encodes the content of the speech. This experiment confirms our observations from before that important features are often those that span many dimensions in the input data space (in this case phoneme and other high-level content in waveform).

We have then analysed the unconditional samples from the model to understand its capabilities. Given the compact and abstract latent representation extracted from the audio, we trained the prior on top of this representation to model the long-term dependencies in the data. For this task we have used a larger dataset of 460 speakers [30] and trained a VQ-VAE model where the resolution of discrete space is 128 times smaller. Next we trained the prior as usual on top of this representation on chunks of 40960 timesteps (2.56 seconds), which yields 320 latent timesteps. While samples drawn from even the best speech models like the original WaveNet [37] sound like babbling, samples from VQ-VAE contain clear words and part-sentences (see samples linked above). We conclude that VQ-VAE was able to model a rudimentary phoneme-level language model in a completely unsupervised fashion from raw audio waveforms.

Next, we attempted the speaker conversion where the latents are extracted from one speaker and then reconstructed through the decoder using a separate speaker id. As can be heard from the samples, the synthesised speech has the same content as the original sample, but with the voice from the second speaker. This experiment again demonstrates that the encoded representation has factored out speaker-specific information: the embeddings not only have the same meaning regardless of details in the waveform, but also across different voice-characteristics.

Finally, in an attempt to better understand the content of the discrete codes we have compared the latents one-to-one with the ground-truth phoneme-sequence (which was not used any way to train the VQ-VAE). With a 128-dimensional discrete space that runs at 25 Hz (encoder downsampling factor of 640), we mapped every of the 128 possible latent values to one of the 41 possible phoneme values¹ (by taking the conditionally most likely phoneme). The accuracy of this 41-way classification was 49.3%, while a random latent space would result in an accuracy of 7.2% (prior most likely phoneme). It is clear that these discrete latent codes obtained in a fully unsupervised way are high-level speech descriptors that are closely related to phonemes.

4.4 Video

For our final experiment we have used the DeepMind Lab [2] environment to train a generative model conditioned on a given action sequence. In Figure 7 we show the initial 6 frames that are input to the model followed by 10 frames that are sampled from VQ-VAE with all actions set to *forward* (top row) and *right* (bottom row). Generation of the video sequence with the VQ-VAE model is done purely in the latent space, z_t without the need to generate the actual images themselves. Each image in the sequence x_t is then created by mapping the latents with a deterministic decoder to the pixel space after all the latents are generated using only the prior model $p(z_1, \dots, z_T)$. Therefore, VQ-VAE can be used to imagine long sequences purely in latent space without resorting to pixel space. It can be seen that the model has learnt to successfully generate a sequence of frames conditioned on given action without any degradation in the visual quality whilst keeping the local geometry correct. For completeness, we trained a model without actions and obtained similar results, not shown due to space constraints.

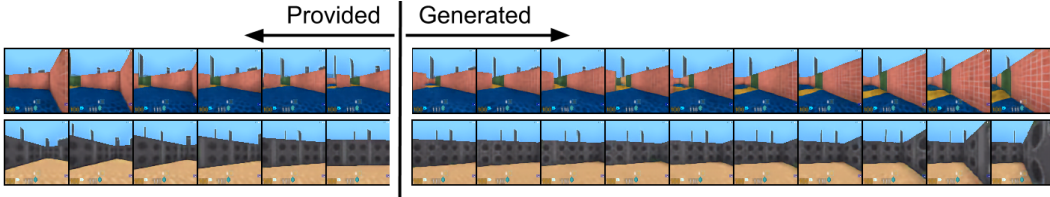


Figure 7: First 6 frames are provided to the model, following frames are generated conditioned on an action. Top: repeated action "move forward", bottom: repeated action "move right".

5 Conclusion

In this work we have introduced VQ-VAE, a new family of models that combine VAEs with vector quantisation to obtain a discrete latent representation. We have shown that VQ-VAEs are capable of modelling very long term dependencies through their compressed discrete latent space which we have demonstrated by generating 128×128 colour images, sampling action conditional video sequences and finally using audio where even an unconditional model can generate surprisingly meaningful chunks of speech and doing speaker conversion. All these experiments demonstrated that the discrete latent space learnt by VQ-VAEs capture important features of the data in a completely unsupervised manner. Moreover, VQ-VAEs achieve likelihoods that are almost as good as their continuous latent variable counterparts on CIFAR10 data. We believe that this is the first discrete latent variable model that can successfully model long range sequences and fully unsupervisedly learn high-level speech descriptors that are closely related to phonemes.

¹Note that the encoder/decoder pairs could make the meaning of every discrete latent depend on previous latents in the sequence, e.g., bi/tri-grams (and thus achieve a higher compression) which means a more advanced mapping to phonemes would result in higher accuracy.

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